Culture Is All You Need: Distributed Network, Antifragile AGI

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Abstract

Current AGI development pursues ever-larger 'single-brain' models, repeatedly re-compressing human knowledge into static weights; a thermodynamically wasteful approach that freezes intelligence and scales entropy faster than intelligence. The fantastic progress gained by scaling laws (3,4,22) demonstrates the immense power of compression, yet faces diminishing returns and inherent limitations; a single model, no matter how large, tends toward dogmatism and uneven performance across contexts. Specifically, the attention mechanisms (5) of single models depend exponentially on scale and data quality; they often abstract away useful contextual knowledge in the pursuit of statistical prevalence, and struggle to learn effectively from 'noisy' or uncurated data, often motivating heavy data curation practices (12). This paper argues that General Intelligence (GI) must transcend static weight models by adaptively tapping into them and other sources of knowledge. We propose a paradigm shift: moving from a monolithic "God-Model" to an evolving Distributed Network Architecture designed for continuous interpretation and contextual adaptation. Rather than specifying a complete implementation, we propose a research paradigm grounded in how distributed intelligence already functions in biological and social systems; from ant colonies to scientific communities to human cultural evolution.

Drawing on the Flynn Effect (1), the documented rise in human IQ scores across generations without genetic change, strongly suggests that intelligence can improve through cultural accumulation, availability, and honing of knowledge, rather than scaling individual capacity. This provides empirical proof-of-concept for distributed, compression-based, evolving architectures. We formalize a framework where agents "tap" into and contribute to a collective reservoir of vectorized skills, strategies, and contexts. To handle the infinite combinatorial complexity of the universe, we define these contexts as 'fuzzy predicates'; partial, probabilistic descriptions that allow agents to act effectively amidst uncertainty without requiring complete state knowledge.

Keywords

Doubt, Context, Strategy, Antifragile, Flynn Effect, Culture, AGI, Compression, Ethics.

Introduction and Motivation

The central challenge of **AGI** is engineering the capacity to define and achieve goals within

diverse, often novel, contexts; "If you don't know where you're going, any road will get you there" (Lewis Carroll) (24); precise goal definition is the prerequisite for effective strategy.

The current "God-Model" paradigm attempts to solve this by compressing all potential contexts into the static weights of a single neural network. While efficient for the statistical mean, this approach falters in the "long tail" of reality. By prioritizing statistical prevalence, monolithic models tend to abstract away the specific, low-probability nuances required for novel contexts, leading to "hallucinations" of the average, and acceptance of common fallacies in the data, rather than precise adaptation.

We submit that a solution lies in shifting the level of analysis from the *individual brain* to *Culture*. The **Flynn Effect** (1,2) provides empirical corroboration that **HGI** is a distributed potential that improves with access to scaled curated and honed knowledge and functionality; rather than biological (hardware) changes.

Our proposal operationalizes this insight. We outline a **Distributed Network Architecture** that integrates LLMs and diverse knowledge sources through a critical, doubt-driven lens. By decoupling **Strategy** from **Actor Intentionality**, we establish a standardized formal tokenization applicable to any form of knowledge or functionality. This enables the system to compress operational wisdom into reusable "Strategy Tokens," circumventing epistemic stances and debates. Complex strategy tokens can be decompressed, recombined, and adapted to new contexts at inference time. **Core Functional Contributions:**

Strategy-Token Graph Calculus: A calculus that represents strategies (e.g., heuristics, hand gestures, empirical testing, logical extrapolation, mathematical modeling) as portable graph tokens composing across all levels of abstraction. Starting from atomic strategies (e.g., axioms, step) that compose by reference into larger graphs, tokens are traversed, recombined, and locally expanded, enabling real-time reuse and adaptation independent of any single agent. Strategy is explicitly decoupled from the actor's intentionality, under Dennett's Engineering Stance (6), the same strategy-token graphs apply to entities as disparate as viruses or mental constructs like "Thor," insofar as attribution yields useful or falsifiable predictions.

Doubt-Triggered Exploration: We formalize Doubt as a contextual measure of predictive uncertainty. High doubt and curiosity trigger exploration; spawning specialized tasks or novel strategies; to escape local optima and discover new knowledge (Active Inference (36)).

Moral Antifragility (The Village of Superegos): We reject paternalistic, monolithic "alignment" filters that rely on data curation (12), post-training deletions, and fine-tuning to sanitize output. As Santayana noted, "Those who cannot remember the past are condemned to repeat it" (15); sanitizing history prevents the system from learning to defend against it. We propose a structural "Moral Space" of diverse, distributed evaluative agents to mitigate malicious intent through collective comprehension and pluralistic debate (9,10).

Formal Framework

This framework assumes descriptions and analysis exist (23), defining concepts in terms of prior concepts. These definitions are intended for their practical utility in AGI, not as formal proofs of origin or independence.

System: A structured set of components and their interactions.The system is described by its state variables, boundary conditions, and regularities explaining its evolution.

System State: A specific configuration of a system's components.

Functions: Descriptions of rules governing change between sets of system states. *These descriptions can be fallacious, inaccurate, heuristic and may not fully capture the complexity of interactions*.

System Subsets: Subsets of a system are smaller, distinct system parts that can be described separately from the larger system.

Objects: descriptions of **system subsets** in a specific **system state**, characterized by exclusive attributes that distinguish them from the adjacent system. They can be distinctly analyzed and interpreted based on these attributes, with varying degrees of persistence and cohesion across states. Theseus's ship is a quintessential example illustrating how an object's distinctiveness is fuzzily described over system states, demonstrating a dynamic and interpretive nature of objecthood. Since **objects** are descriptions they may abstract and describe multiple **system states.** *A platonic object is an immutable description*.

Idiosyncratic changes: Functions that almost exclusively affect the altered **system subset** itself within a defined scope of **system states**. *For example the flapping of a butterfly's wings affects the trajectory of the butterfly*.

Systemic changes: Functions that have significant effects on other subsets of the system within a defined scope of system states. *For example the flapping of a butterfly's wings affects the trajectory of the butterfly into the eye of the driver of the president of the United States, causing his death and world war 3*.

Network: A **System Subset** composed of an interconnected structure of nodes (**objects**) and edges (**functions)** that describe the relations between objects across **System States.**

Networks can be described and analyzed as objects.

Agent: Analysis of an **object,** sufficient for attributing to it a set of **functions** which influence the probability of a definable set of **system subset** configurations. Agents vary in cohesiveness and distinction from other parts of the system.

Actions: Agent functions.

Agent Steps Graph: Nodes are **agent states**, and edges are **actions**. A finite graph consisting of nodes and directed edges, where each edge has a direction, and optional cycles are finite.

Model: a description of a **system subset** or possible subset.

Analyzing Agents: Agents capable of **Modeling agents, networks**, and **actions** and predicting resulting system states (n≥1). Predictions and models may be erroneous or vary in degrees of accuracy, completeness and utility.

Origin State: The **system state** from which an agent is analyzed or a **model** of it.

Goals: Subsets of **system states** that "**analyzing agents"** assign to "**acting agents**" under analysis.

Strategy: The evaluation of **agent actions** by an **analyzing agent**, in relation to the probability of transitioning from an **origin state** to a set of **goals** (n≥1) the analyzing **agent** attributes to **acting agents**, within a defined or implied **system**. Assessing probability may involve varying degrees of confidence and knowledge. *An analysis may be erroneous, for example, potential categorical mistakes in goal attributing, such as regarding people's eating habits solely in relation to the goal of achieving certain conscious weight goals rather than a subconscious evolutionary heuristic, adaptive to situations with scarce resources.*

The decoupling of the analyzing **agent** from the acting **agent**, allows coherent talk about **strategy** without necessarily attributing the **agent** the ability to strategize. Thus one can coherently speak of the **strategy** of a virus (18), without attributing to it **modeling**, explicit **goals** and planning. The above definition of **strategy is novel** in that it fully **decouples agenthood from any requirement of stable identity or explicit intentionality.** Instead of limiting "strategy" to a well-defined agent (28), player (31)(32) or organism (29)(30), the framework allows **any subset of a system**, even an ephemeral abstraction like "democracy" or "Thor", to be attributed a set of goals and actions by an external analyzer. This breaks away from traditional definitions of strategy allowing dissolution of classical boundaries between agent and observer, permits partial or erroneous attributions, and treats "strategy" as a *useful modeling stance* rather than a strict property of a self-contained, intentional entity.

Light Cone of Consequences: A finite set of N functionally connected consecutive **system states** affected by a given **action** or **strategy**. Light cones help constrain the predictive scope of agent strategies, defining practical bounds for computation and analysis.

Intentional Agents: Analyzing agents that model their own **goals** and actions; levels of awareness, modeling, and control may vary from being completely erroneous to comprehensive (e.g., knowing all moves in Tic Tac Toe).

Rational Agents: Rational agents are intentional agents that assess the probability of their **strategies** achieving their **goals**; in vast or boundless systems, finite agents' modeling may be bound by information fidelity and size. Rational analysis is bound by **Goals** i.e. it is meaningless to assign probability without an effect.

Culture: A **network** of **agents** and their strategies that interact in ways that give rise to a meta entity (16), which can be analyzed holistically in terms of **strategy**. A beehive contains no "queen mind," yet the colony regulates temperature, builds geometry, and reallocates labor through distributed signaling; that can be analyzed in terms of colony survival and efficiency.

Predicate: A statement pointing from a set of **system subsets** to a set of **system subsets**. If the origin or pointed subset are not provided completely, different predicates may look equivalent.

Context: A **system subset** which corresponds to a **predicate** e.g. chemistry or chess. Some contexts are predicated completely, such as Euclidean geometry and predicate calculus, while others are unclear and heuristic like love.

Fuzzy Context: A context predicated with varying degrees of truth, rather than a binary evaluation. It is characterized by a membership function that assigns a truth value between 0 and 1 to each element in its context/domain, indicating the degree to which the element satisfies the predicate.

Distribution of Contexts: The combinatorial set of all possible subsets of the system. Each context represents a unique combination of components, interactions, and states within the system. ***Contexts*** can overlap or map to **agentic goals**.

Context Compression: Small representation of large contexts.

Chunking: A sparse representation of **contexts e.g. gardening and laying bricks chunk** to manual Labor.

Context Decompression or Parcelling: breaking down a complex **goal** into smaller, manageable **goals**.

Parcelling and **Chunking** are useful for optimizing **Agent Steps Graphs** aligned with **context** expert agents. When a task is complex one can reduce doubt by dividing it and assigning parts to **domain** expert **agents**. When aligning an agent with a task it is often helpful to shorten search by narrowing categories or ***Chunks.***

Exploit: Agent reuse of functionality and knowledge e.g. **Agent Steps Graphs, strategies, contexts, Context Compression**, or **models**.

Explore: Agent **strategies** involving novel actions, novel combinations of **Agent Steps**

Graphs, compressions, decompressions, contexts or models.

Hypotheses: models of unknown system states.

Success: The degree to which a strategy instantiates the **goal** assigned to it in analysis.

Skill: exploit of stored **successful strategies**, or agent graphs **optimized for specific** goals.

Culture acts as a medium for storing, transmitting, refining and inventing **skills**, reducing the need for agents to "retrain" from scratch. Systems like language, mathematics, text, code and rituals serve as repositories of compressed **skills**, enabling agents to access and reuse knowledge efficiently.

Explanation: A **strategy** with the **goal** of transferring **skill** between agents. *Future refinements may explore subtypes of **Explanation***.

General Intelligence (GI): The ability of an agent to form goals and **successfully** achieve them, by creating, recombining and applying **strategies** in many diverse **contexts**. ***GI*** includes: Acquiring, applying, repurposing, and inventing **skills**.

Heuristic: A **strategy** prioritizing **success** in a set of **contexts** over **model** accuracy (17). Heuristics use simplified rules or approximations to achieve **goals** under resource constraints. While suboptimal in some cases, they are statistically beneficial in **context**.

*A cat crossing a street may use the heuristic: **"If it's smaller or slower than a wolf, it's safe to go."** This rule enables quick decisions but fails for cars; cats that use a generalized fear **heuristic** can adapt and survive in urban car infested **contexts**.*

Tactic: A **skill**-intensive **action** analyzed as a sub-**goal** within a larger **context**. The same action can be analyzed in relation to its **goal** as a **strategy** or **tactic**, depending on the analysis point of view. *For example, the "Taking of Omaha Beach" on D-Day can be analyzed as: A **strategy**, from the perspective of Lieutenant General Omar Bradley (Commander of the U.S. First Army), whose **goal** was to establish a beachhead in Normandy. A **tactic**, from the perspective of General Dwight D. Eisenhower (Supreme Commander of the Allied Expeditionary Force (SCAEF) for Operation Overlord), whose broader **goal** was to liberate northwest Europe from German occupation. A reusable **skill**-intensive **action**, from the perspective of Major General Clarence R. Huebner (Commander of the U.S. 1st Infantry Division), who ensured his troops were well-drilled in amphibious landings, platoon-level warfare, and infantry skills, which could have been deployed in Italy, Norway, or Zimbabwe.*

Determining Intentionality: In **contexts** subject to computational irreducibility (25), confusion regarding whether an **agent's goals** are intrinsic or attributed by an observer is often unavoidable, forcing analysis to rely on probabilistic attributions and accept potential misalignment until further information is obtained.

Engineering Stance: As long as assigning agency and **goals** lead to testable refutable predictions they are useful hypothesis generating tools i.e. they provide a scheme for **explorative** strategies.

Risk: A measure of the probability that the system's state, within a defined **light cone**, does not belong to the **goal** set, typically expressed as 1 - Success.*This captures the uncertainty under computational irreducibility, reflecting how multiple possible outcomes may fall outside the **goal** set.*

Doubt: A subjective assessment of **risk** by the analyzing agent. Estimation of the likelihood that its **model** is false or the **strategy** may fail i.e. that the system's state could lie outside the goal set, reflecting the nonzero risk (1 - **Success**) under computational irreducibility. **Doubt** arises from the uncertainty embedded in the analyzing agent's knowledge and models, acknowledging that risk remains whenever success is less than certain.

A note about Open Endedness: Having a goal is considered by some (33) an antithesis to open endedness, on the contrary I would argue that **Open Endedness** is about the **goal** of **exploring**, looking for **goals**, pursuing some, finding more and switching pursuit. Pursuing a **goal** is a continuum between a fleeting capture of attention, and an all consuming pursuit.

The apparent Computational Irreducibility (25) of our universe(s), provides seemingly infinite spaces to explore, **points of view**, **goals** and **strategies** to pursue. The mechanisms for deciding which **goal** to switch to, and how easily, are most probably varied and complex, perhaps infinitely so. The tendency to choose **strategies** of varying **goal** switching magnitudes, are in themselves meta **strategies**, in natural language they have many names and conceptualizations, such as various temperaments and personalities.

Open Endedness: A novel **goal** seeking **strategy**. Open Endedness *is a high risk strategy considering finite resources, it holds potential of gains and loss. An agent would have to have basic redundancy to survive pursuing novelty in a manner akin to the Maslow hierarchy of needs (34); One could conceptualize the willingness of adventurous types to risk doom by analyzing them as parts of larger entities, such as family, tribe, nation, species.*

Dogmatism: Low probability of switching **goals** and **agent steps graphs**. Choosing to write about open endedness before defining it, is a conscious departure from the dogma of the first part of the framework. **Wondering:** High probability of switching **goals** and **agent steps graphs. Walkabout:** Traditional aboriginal **wondering**.

Adventure: operating in unknown spaces, novel contexts or strategies. Adventure yields high information, skill acquisition and increased possibility of loss, unless the **origin system state** holds more risk. Ossification of culture due to lack of exploration and adventure can be maladaptive.

Architecture:

The core of the architecture is the efficient storage, retrieval and realtime mutability of **Strategy Tokens in Vector Spaces** .We define four primary vector spaces that allow agents to compress, retrieve, and evolve knowledge.

Strategy Doubt Space: A vector space where tasks are mapped to strategies (27). Simple tasks map to a single Basic Agent Step; complex tasks are **parceled** (11) into hierarchies or **chunked** into domain coordinators. **Vector Components: Goal** (Pointer to System State); **Agent Steps Graph** (The recipe for action); **Performance Score** (Success metrics); **Scope & Alignment** (Degree to which output accomplished the goal).

Context Doubt Space: A vector space where strategies are mapped to **contexts** (predicates). This space helps agents parcel tasks and assign specialist agents.**Vector Components: Strategy** (Origin State, Graph, Goal); **Context (Predicate)** (Fully or partially defined fuzzy predicate); **Success Metrics** (Effectiveness in *that specific context*); **Similarity in Plans**.

Task Utility Space: A vector space where the effectiveness of tasks is scored based on generalizability and systemic learning effects. **Vector Components: Specificity** (Fit to context); **Generalizability** (Applicability to other tasks); **Systemic Effects** (Enhancement of learning across contexts); **Bias Potential**.

Moral Doubt Space: A vector space where the moral evaluation of tasks is stored. **Vector Components: Moral Evaluation** (Assessment of goals/sub-goals by humans or agents); **Light Cone of Consequences** (Evaluation of idiosyncratic vs. systemic effects up to acceptable scope); **Law Abidance; Contextual Relevance**.

Evolutionary Mechanisms: We use a **Stochastic Genetic Evolutionary Algorithm** with **self-play** (19,10) to search strategy latent space. Agents are randomly assigned to tasks, evaluated via the Strategy Doubt Space, and the most successful graphs are selected for **crossover** (recombination) and **mutation** (random changes)reducing stagnation in local contextual optima. We evolve strategies under selection pressure; LLMs can serve as adapters between heterogeneous tools so interfaces can change without global refactors. Crucially, we incorporate **Failure Tolerance**: novel strategies are periodically reintroduced and protected from premature elimination in irregular domains.

Curiosity: A "Doubt Agent" monitors contexts for low coverage, isolation, or stagnation spawning tasks to improve them; implements reflective and epistemic update patterns (20) alongside Active Inference (36). By selecting epistemic policies that minimize expected free energy under low predictive precision, we reduce and actively fix vulnerability to rare or distribution-shifting events and low information coverage. Example: Directive: fix torn pants in the hamper. Robot plan: (1) go to hamper (2) retrieve damaged pants (3) go to sewing kit (4) fix pants (5) report by text. Execution: real-time nav-to-(hamper, cached map) → locate(pants) → bend → grasp. (2) nav-to-(cached map, bedroom closet) → retrieve(sewing

kit). (3) parcel “fix” into a sewing subgraph (thread → align → stitch) → fold → place in closet → text user → submit update: select smaller thread for dry-fit fabric to sewing strategy for future reuse.

Discussion and Conclusion

Moral antifragility carries the risk of gridlock; however, human systems such as Halachic law, democracy, and markets suggest that structured pluralism is often more adaptive than imposed uniformity. Unlike alignment approaches based on aggressive curation or post-training deletion (13), the Moral Doubt Space models and flags malicious intent to mitigate it. IRL and preference-based alignment (7,8) are powerful for distilling latent strategies from human behavior into reusable strategy tokens. Embedding inferred reward structures directly into static weights reduces the probability of goal and strategy switching across contexts; rendering the system dogmatic in the sense defined above. Validation may include scenario-based moral dilemmas (opinion spectrum coverage and pluralism) and longitudinal ethical consistency (stability under perturbation (26)). We embed ethics, doubt, conservation, curiosity, and exploration directly into the architecture; distributed competence rather than post-hoc curation. "Love thy fellow as yourself; the rest go out and learn" (Hillel) (21).

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